

THIRUVANANTHAPURAM, India

WMO: 433710

Lat: 8.508N

Lon: 76.958E

Elev: 210

StdP: 14.58

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	72.1	73.1	61.7	82.9	83.6	64.1	90.2	82.4	15.3	83.7	13.5	84.2	2.0	50	0.312

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.9	93.6	79.2	92.5	78.8	91.5	78.4	82.5	89.5	81.7	88.7	81.0	87.9	4.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	160.2	86.8	79.8	156.0	86.0	79.1	152.4	85.3	46.5	89.7	45.7	89.0	44.9	88.3	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.4	7.7	68.5	96.7	3.2	2.5	66.2	98.5	64.4	100.0	62.6	101.4	60.3	103.2		
(5)			64.2	84.3	1.9	1.4	62.8	85.3	61.7	86.1	60.6	86.9	59.2	87.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.1	81.7	83.0	84.6	84.8	84.0	81.3	80.5	80.8	81.3	81.2	81.1	81.4	(6)
(7)		DBStd	2.35	1.51	1.52	1.68	2.11	2.38	2.22	1.80	1.77	1.74	1.90	1.67	1.57	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11734	982	923	1072	1044	1054	940	946	956	940	969	934	974	(10)
(11)		CDD65	6259	517	503	607	594	589	490	481	491	490	504	484	509	(11)
(12)		CDH74	64904	5205	5504	7308	7282	7000	4804	4335	4554	4695	4708	4518	4991	(12)
(13)		CDH80	23559	2035	2285	3256	3299	2915	1461	1136	1252	1395	1410	1369	1746	(13)
(14)	Wind	WSAvg	2.6	1.6	1.9	2.2	2.5	3.1	3.5	4.4	4.3	3.3	2.0	1.4	1.4	(14)
(15)	Precipitation	PrecAvg	70.1	0.7	0.9	1.5	4.7	8.0	11.9	8.0	5.9	7.1	10.9	8.1	2.5	(15)
(16)		PrecMax	95.3	6.4	5.0	12.3	12.2	23.3	34.3	16.6	20.7	19.6	25.8	28.4	13.5	(16)
(17)		PrecMin	44.4	0.0	0.0	0.0	0.2	0.3	2.3	1.9	0.8	0.2	0.7	0.6	0.0	(17)
(18)		PrecStd	12.6	1.2	1.3	1.9	2.7	5.0	6.6	3.5	3.9	5.4	5.7	5.1	2.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.9	93.6	95.0	95.2	93.8	91.5	89.1	89.7	90.8	91.1	90.6	91.4	(19)
MCWB			74.9	75.6	78.5	80.6	81.3	79.8	79.4	78.3	78.4	78.0	78.6	76.5	(20)	
2%		DB	90.8	92.1	93.5	93.5	92.2	89.3	87.8	88.2	89.2	89.5	89.3	90.0	(21)	
		MCWB	75.2	75.5	78.5	80.6	80.8	80.0	78.8	78.3	78.5	78.2	78.2	76.5	(22)	
5%		DB	89.7	90.9	92.3	92.3	91.0	88.0	86.7	87.1	88.0	88.1	88.1	89.0	(23)	
		MCWB	75.2	75.5	78.3	80.3	80.5	79.6	78.6	78.2	78.3	78.4	78.1	76.5	(24)	
10%		DB	88.6	89.8	91.0	91.0	89.7	86.7	85.5	85.9	86.7	86.7	86.8	87.8	(25)	
		MCWB	75.3	75.8	78.1	80.1	80.3	79.2	78.3	78.1	78.1	78.3	77.9	76.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.9	80.1	82.6	83.6	82.4	81.0	80.7	81.8	81.3	81.0	80.8	(27)	
MCD B			88.0	89.1	90.7	91.1	90.8	87.7	86.6	86.2	86.9	86.8	87.4	87.6	(28)	
2%		WB	78.5	79.1	81.0	82.3	82.3	81.0	80.0	79.7	79.9	80.0	79.9	79.5	(29)	
		MCD B	86.3	87.7	89.5	90.0	89.3	86.6	85.5	85.4	85.9	86.1	86.4	86.4	(30)	
5%		WB	77.7	78.3	80.1	81.6	81.6	80.3	79.2	79.0	79.1	79.3	79.2	78.6	(31)	
		MCD B	85.4	86.6	88.6	89.2	88.3	85.7	84.8	84.9	85.4	85.3	85.5	85.5	(32)	
10%		WB	77.0	77.6	79.5	80.9	80.9	79.7	78.6	78.3	78.4	78.7	78.5	77.9	(33)	
		MCD B	84.6	85.5	87.8	88.5	87.4	85.0	84.1	84.0	84.5	84.6	84.5	84.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.1	15.0	14.2	12.9	11.3	10.1	10.0	10.3	11.0	11.3	11.6	13.5	(35)
MCDBR			16.0	15.9	15.1	14.1	12.5	11.7	11.4	11.7	12.6	13.0	13.5	15.1	(36)	
MCWBR			6.1	5.5	5.1	5.1	4.6	4.9	4.8	4.4	4.6	4.8	5.3	5.9	(37)	
5% WB		MCDBR	14.3	14.1	14.1	13.1	11.8	10.6	10.5	10.7	11.3	11.7	12.1	13.0	(38)	
		MCWBR	5.8	5.1	5.2	5.6	5.2	5.1	5.0	4.7	4.9	5.2	5.5	5.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.475	0.499	0.568	0.577	0.577	0.530	0.549	0.525	0.483	0.518	0.528	0.496	(40)	
taud		2.196	2.123	1.953	1.953	1.960	2.103	2.030	2.112	2.249	2.136	2.086	2.156	(41)		
Ebn at Noon		266	263	246	241	236	244	240	251	265	255	249	257	(42)		
Edh at Noon		47	51	62	61	59	50	55	51	45	50	52	48	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1730	1944	2004	1892	1729	1586	1621	1712	1783	1619	1402	1511	(44)	
(45)		RadStd	81	77	117	93	137	129	107	119	123	108	90	95	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.50	N/A	N/A	+0.99	+1.08	+0.99	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.50	N/A	N/A	+0.99	+1.08	+0.99	N/A	N/A	N/A	N/A

Nomenclature: See separate page